



# Nonparametric Analysis of Feature-Driven Performance in Wearable Health Devices

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# Dataset description

Our dataset provides a comprehensive analysis of the performance metrics of leading wearable health devices. It covers a wide range of popular **brands**, **models** and **categories** (smartwatches, fitness trackers, smart rings, fitness bands, and sports watches).

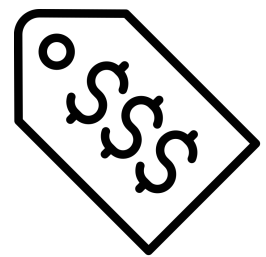


# Research questions

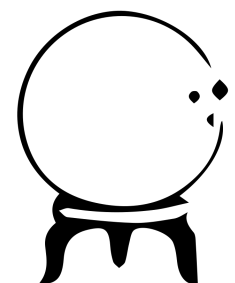
## Stakeholder: customer interested in buying a wearable device



Which device category offers the best balance of accuracy, battery life and other features for typical users?



Is a higher price justified by better accuracy and functionality or just by the brand's popularity?



Can we predict how a new device would perform in terms of user satisfaction?

# RQ1: Best Category

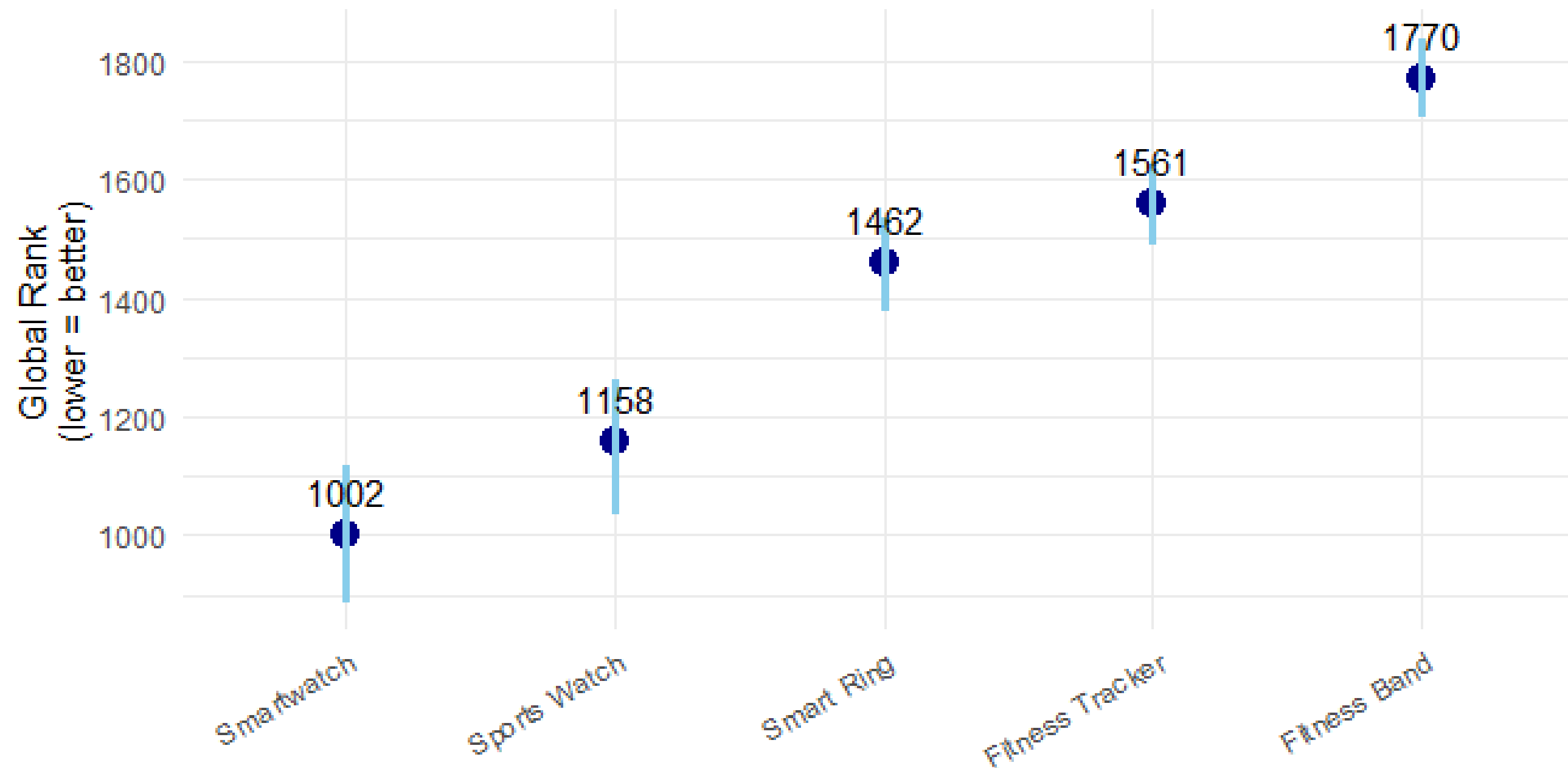
**Compute device-level rankings:** each device is ranked on multiple performance dimensions (higher performance = lower rank);

**Build a global Rank:** all nine metric ranks are averaged to produce a single composite score representing overall performance;

**Global significance test (Permutational ANOVA):** **Category** is a significant factor;

# RQ1: Best Category

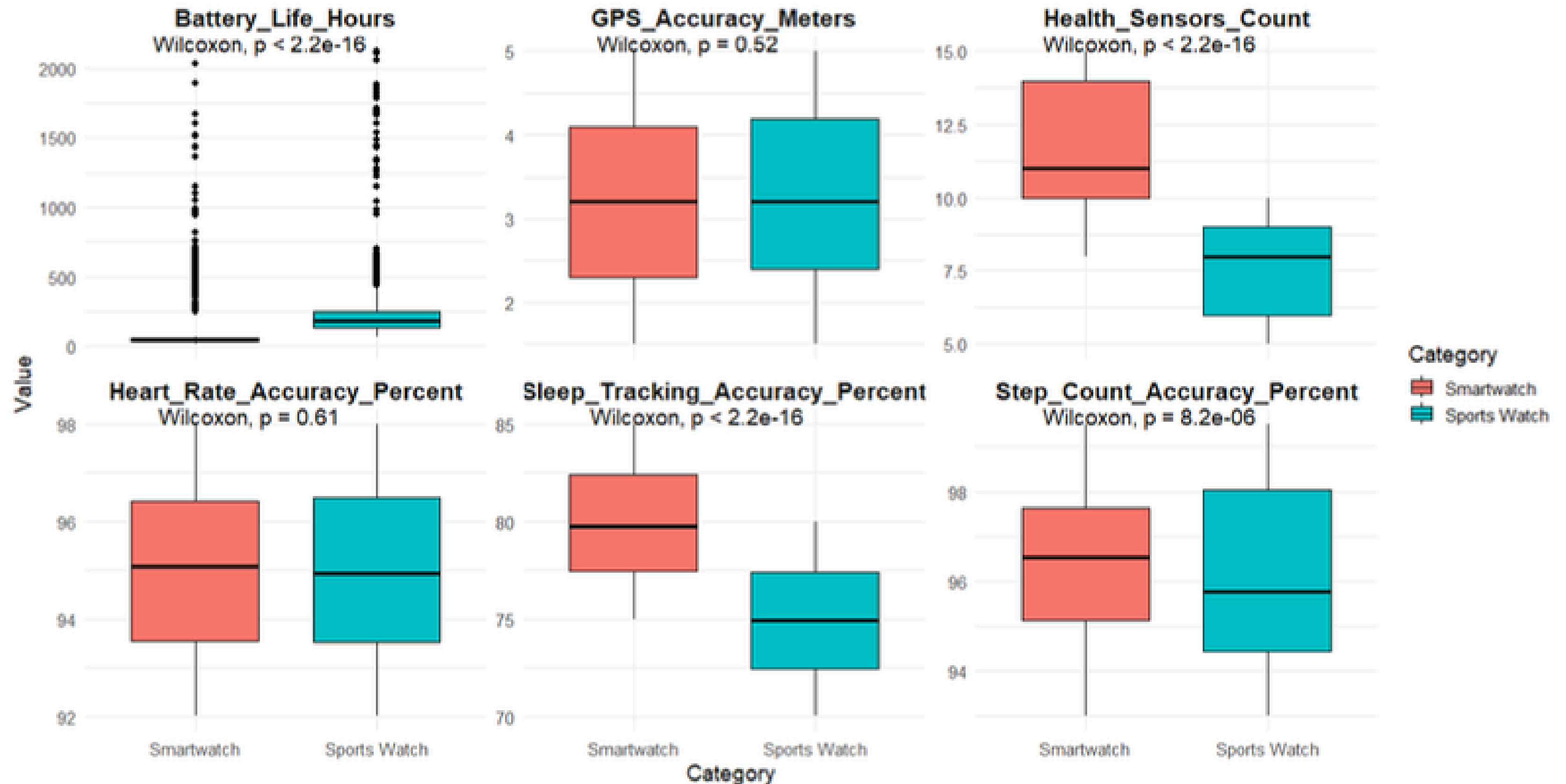
**Pairwise comparisons:** Permutation test comparing median global Rank between every pair of categories.



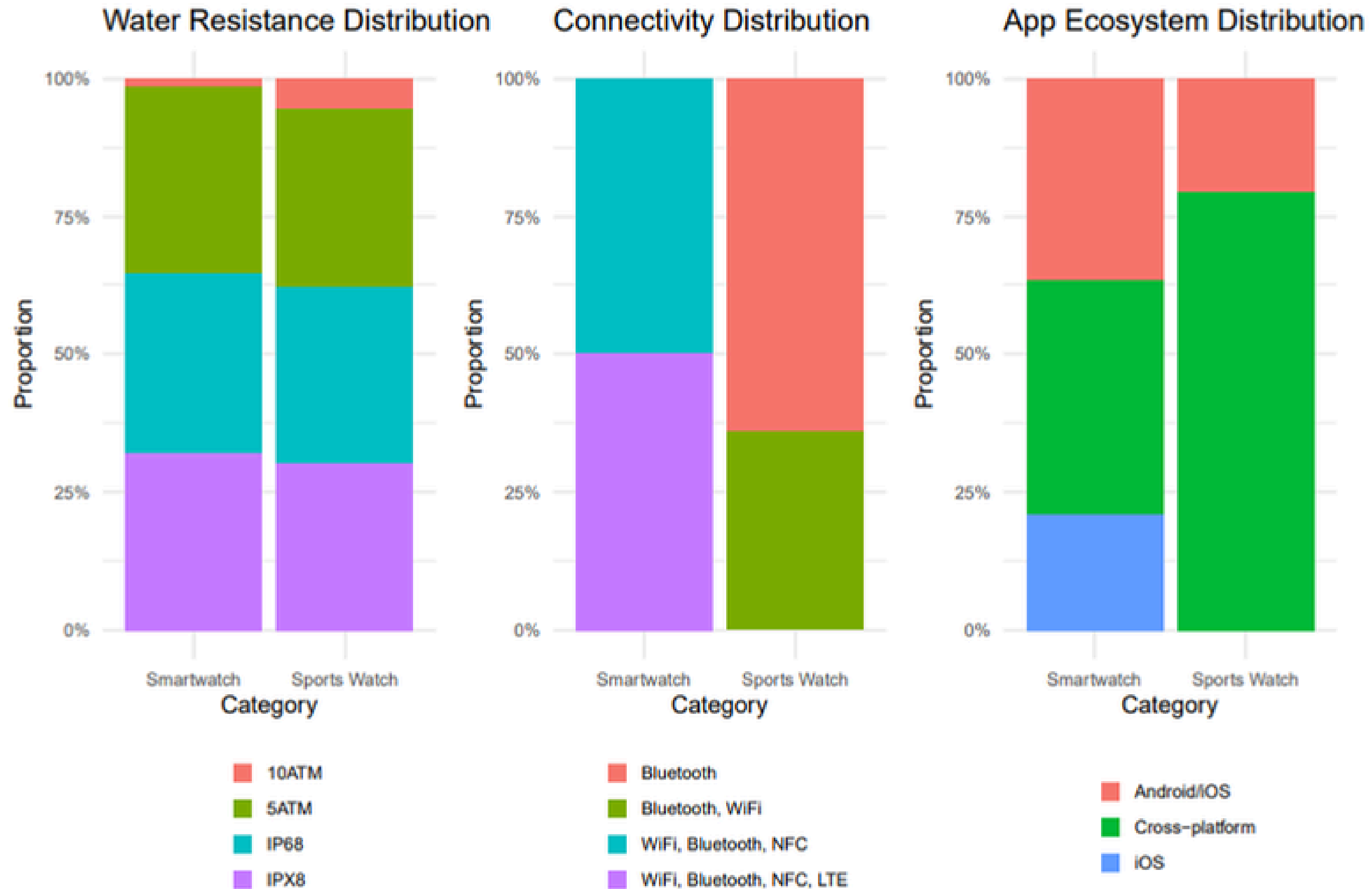
The best results are obtained by **Smartwatches**.



# Comparing the two best categories



# Comparing the two best categories



## RQ2: What justifies the price?

**Normalized rank scores:** We transformed all performance-related variables into normalized rank scores;

**Accuracy and Functionality indexes:** the normalized rank scores were aggregated into two components: an Accuracy Index and a Functionality Index;

**Performance index:** The two components were combined with equal weights into a single Performance Index, representing an overall measure of technical quality.



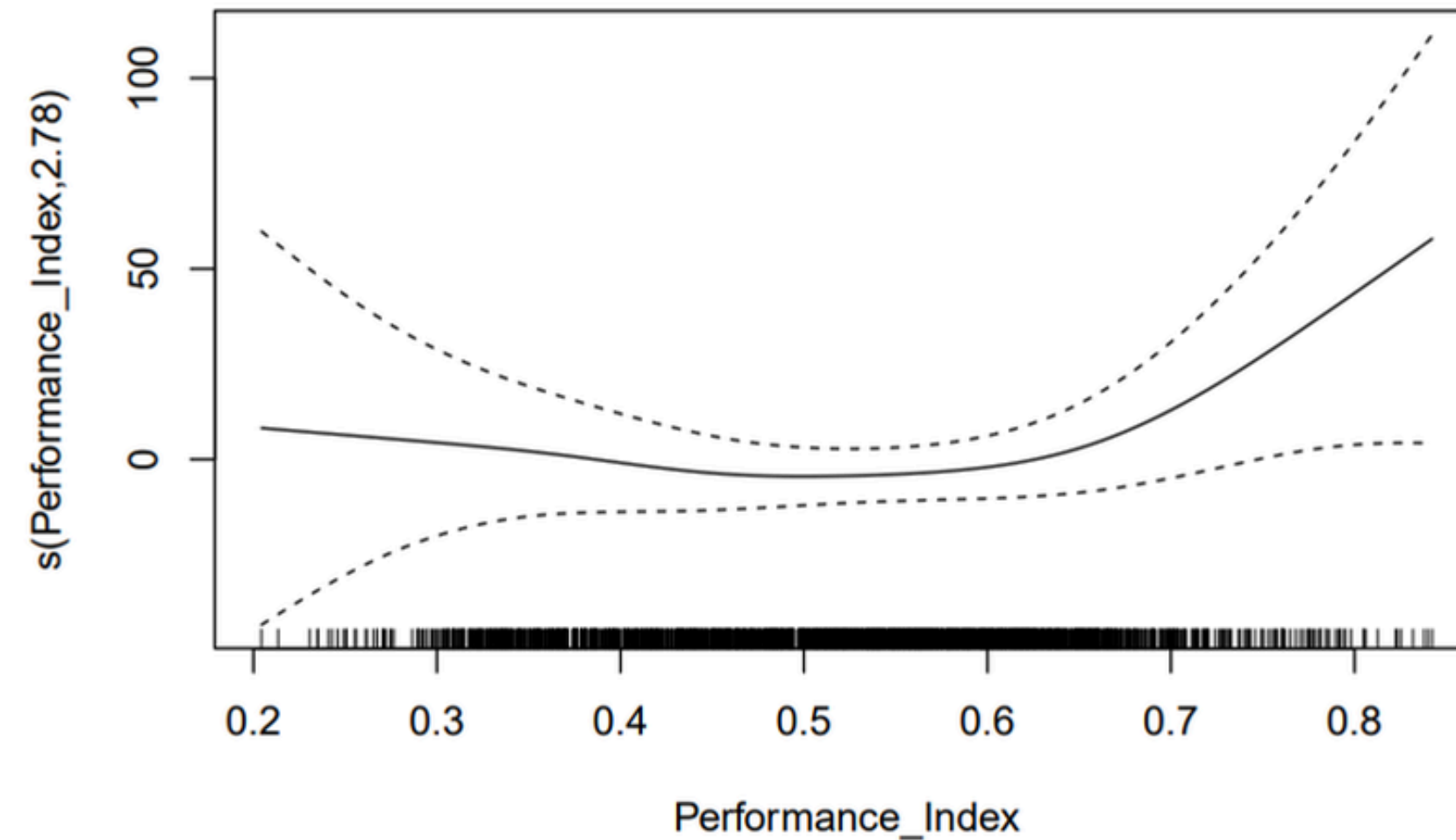
## RQ2: What justifies the price?

**GAM:** price as a function of the constructed Performance Index and brand identity for the  $i$ -th device.

$$\text{Price}_i = \beta_0 + f(\text{Performance Index}_i) + \sum_{b=1}^8 \beta_b I\{\text{Brand}_i = b\} + \varepsilon_i$$

**Results:** the smooth effect of the Performance Index is not statistically significant, whereas brand effects are large and highly significant for almost all manufacturers. Nearly 44% of the price variability is explained.

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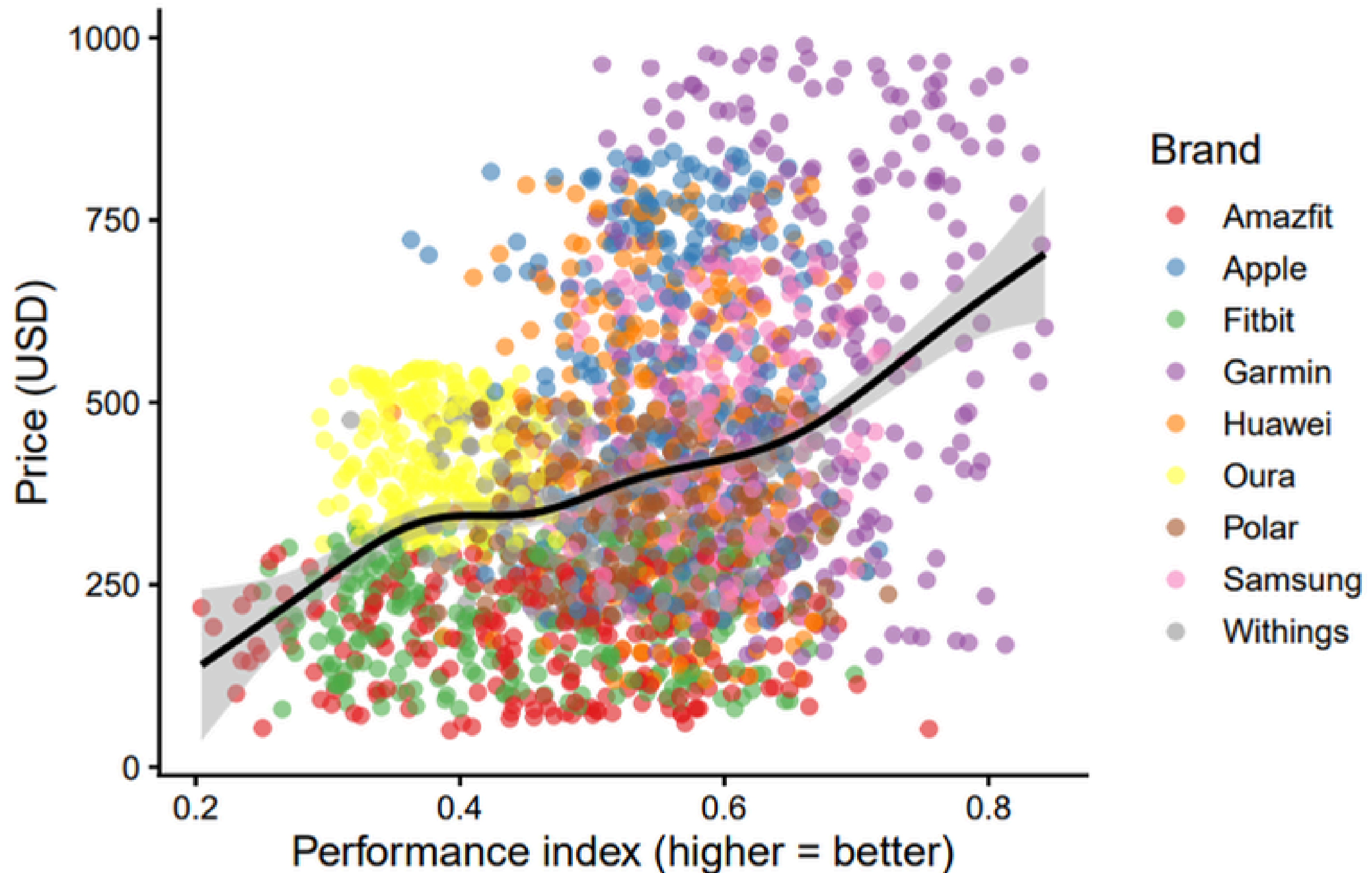
## RQ2: What justifies the price?

**Isolate Performance Index:** GAM without brand effects for the i-th device.

$$\text{Price}_i = \beta_0 + f(\text{Performance Index}_i) + \varepsilon_i$$

Performance Index exhibits a strong and highly significant nonlinear association with price, but the proportion of explained variability drops to around 12%.

## RQ2: What justifies the price?



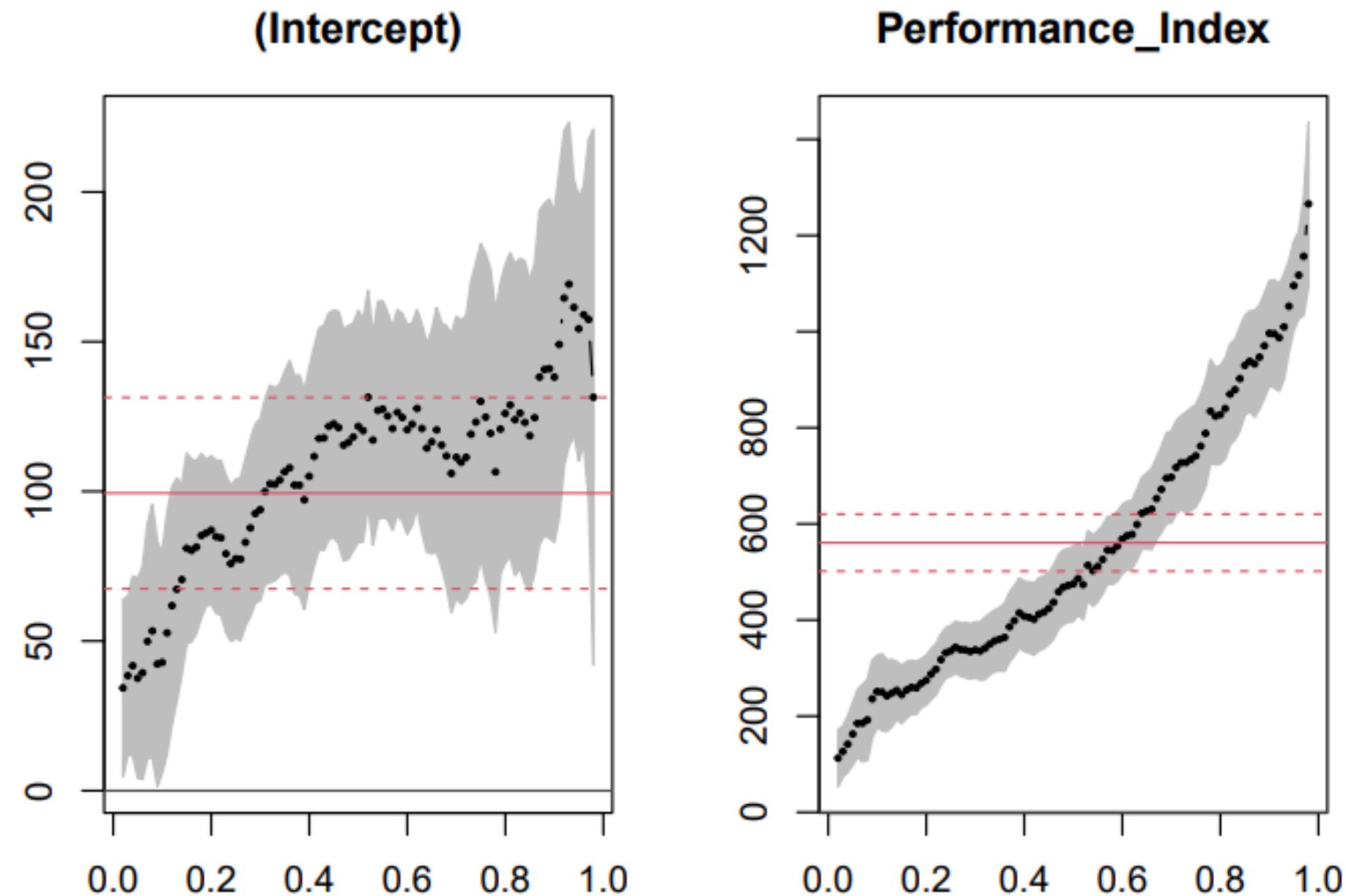
## RQ2: What justifies the price?

**Results:** while a “baseline” performance-price relationship exists, the vast majority of the price premium is dictated by brand positioning rather than technical specifications.

To further investigate the relationship between technical performance and price, we complemented the mean-based analysis with a **quantile regression** approach.

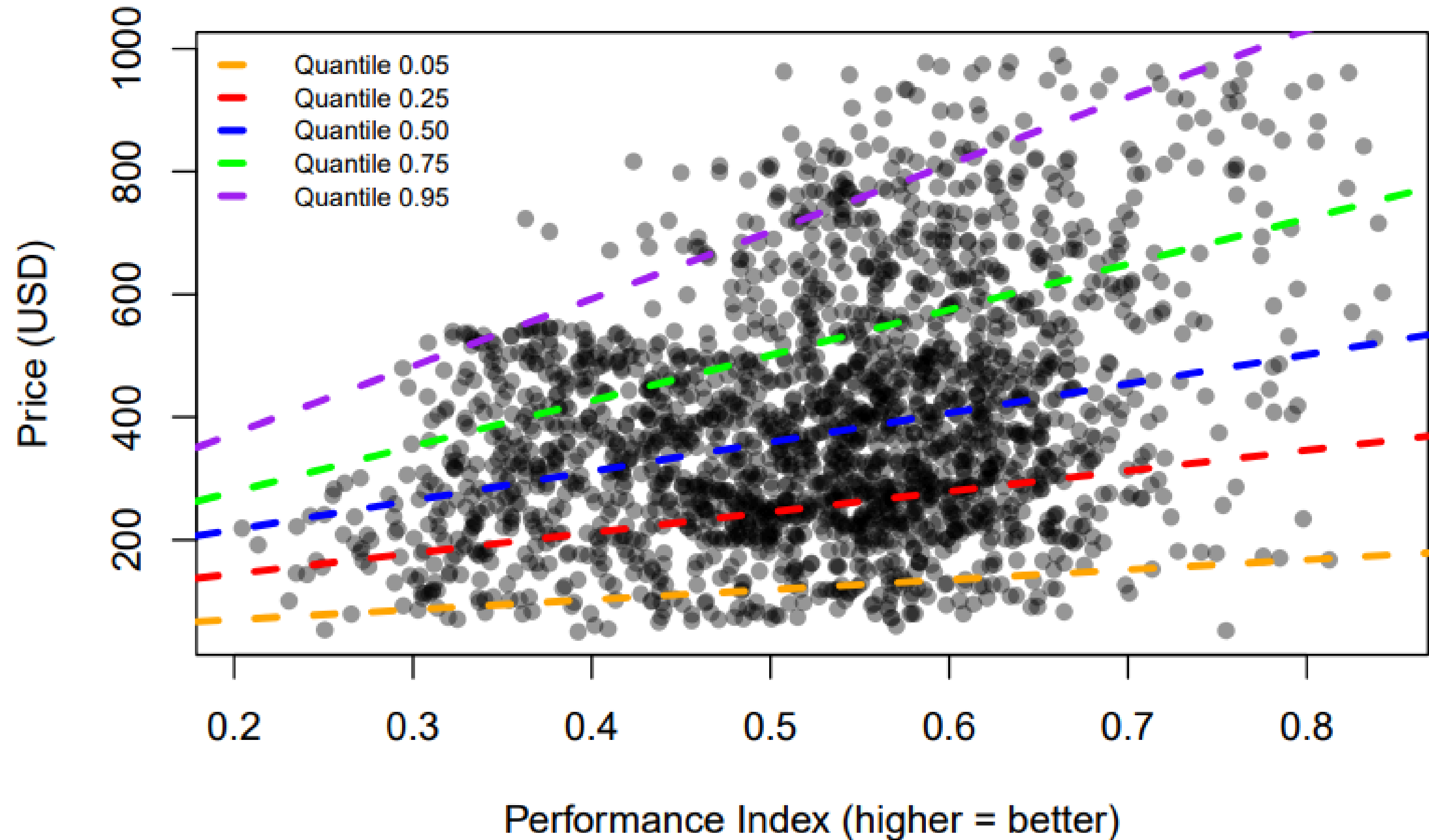


## RQ2: What justifies the price?

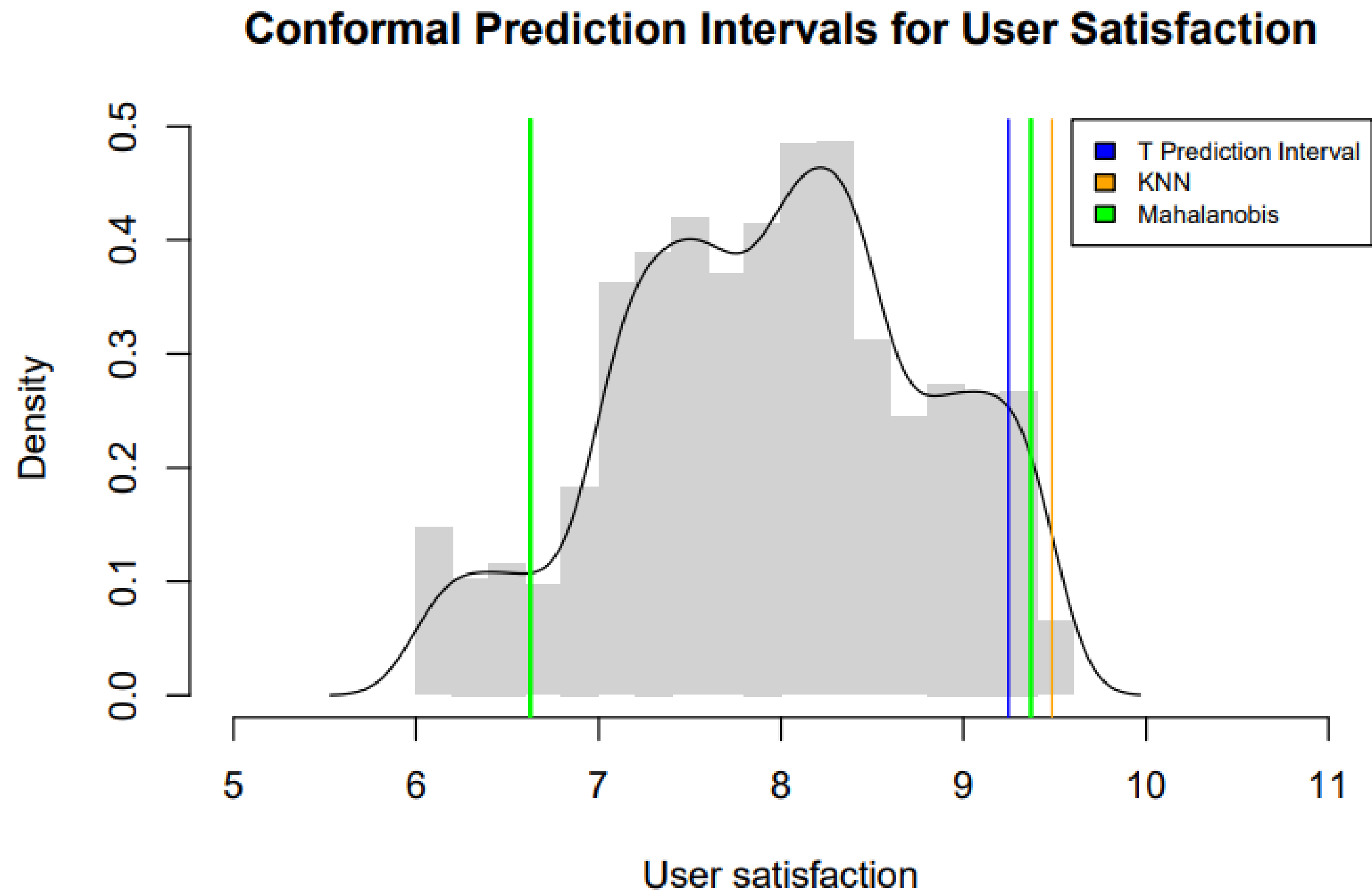


The **fitted quantile coefficients** reveal substantial variation in slopes across quantiles: the association between performance and price is not constant.

## RQ2: What justifies the price?



# RQ3: Predicting user satisfaction





## RQ3: Predicting user satisfaction

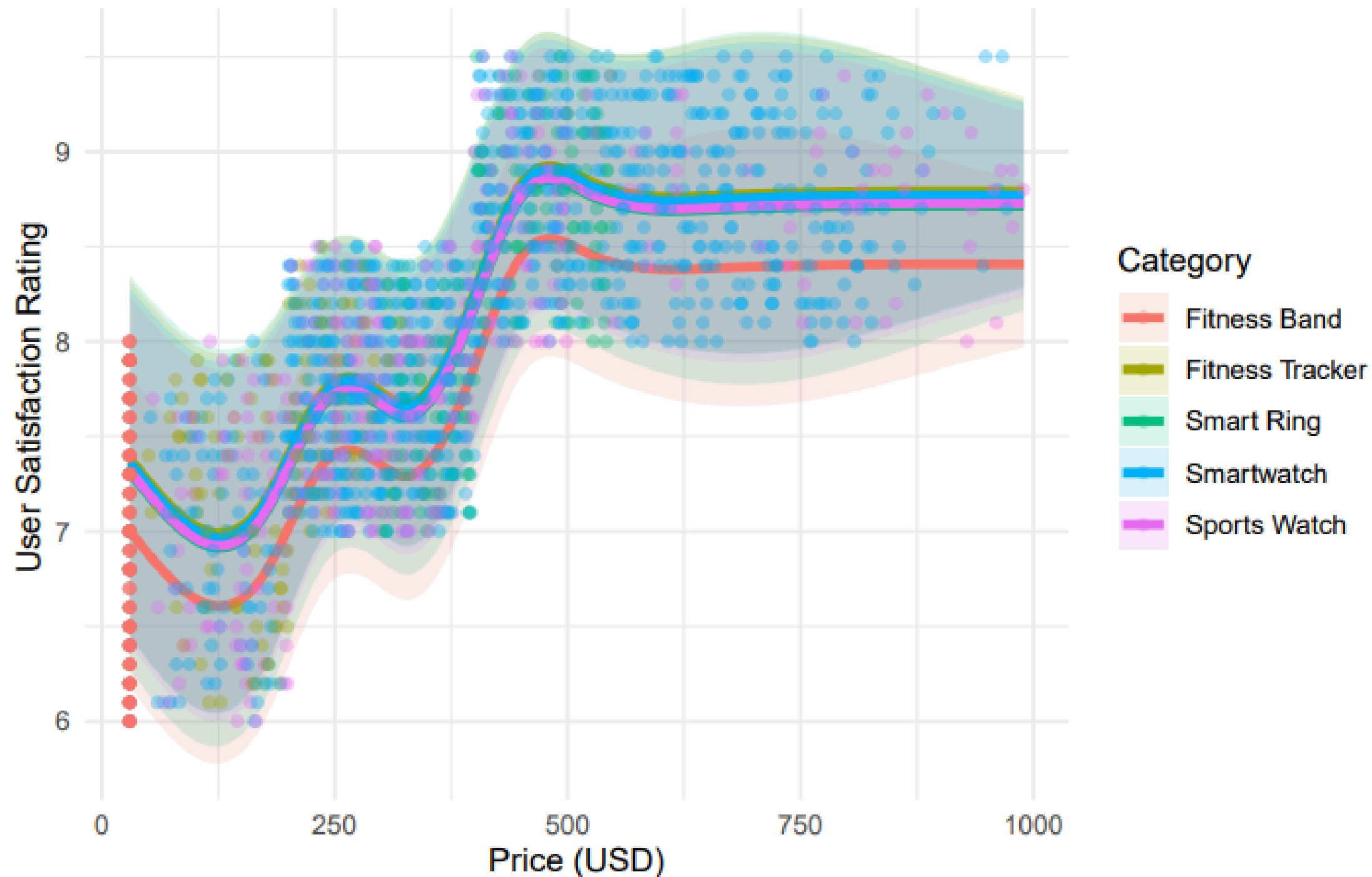
**Splitting the dataset:** we first split the dataset into a training set (75%) and a calibration set (25%);

**From full GAM to final model:** after initially including all technical, market and brand covariates, we performed a sequence of permutation-based significance tests, using the Freedman and Lane scheme, to assess the contribution of each predictor.

$$\text{User Satisfaction}_i = \beta_0 + f(\text{Price}_i) + \sum_{b=1}^4 \beta_b I\{\text{Category}_i = b\} + \varepsilon_i$$

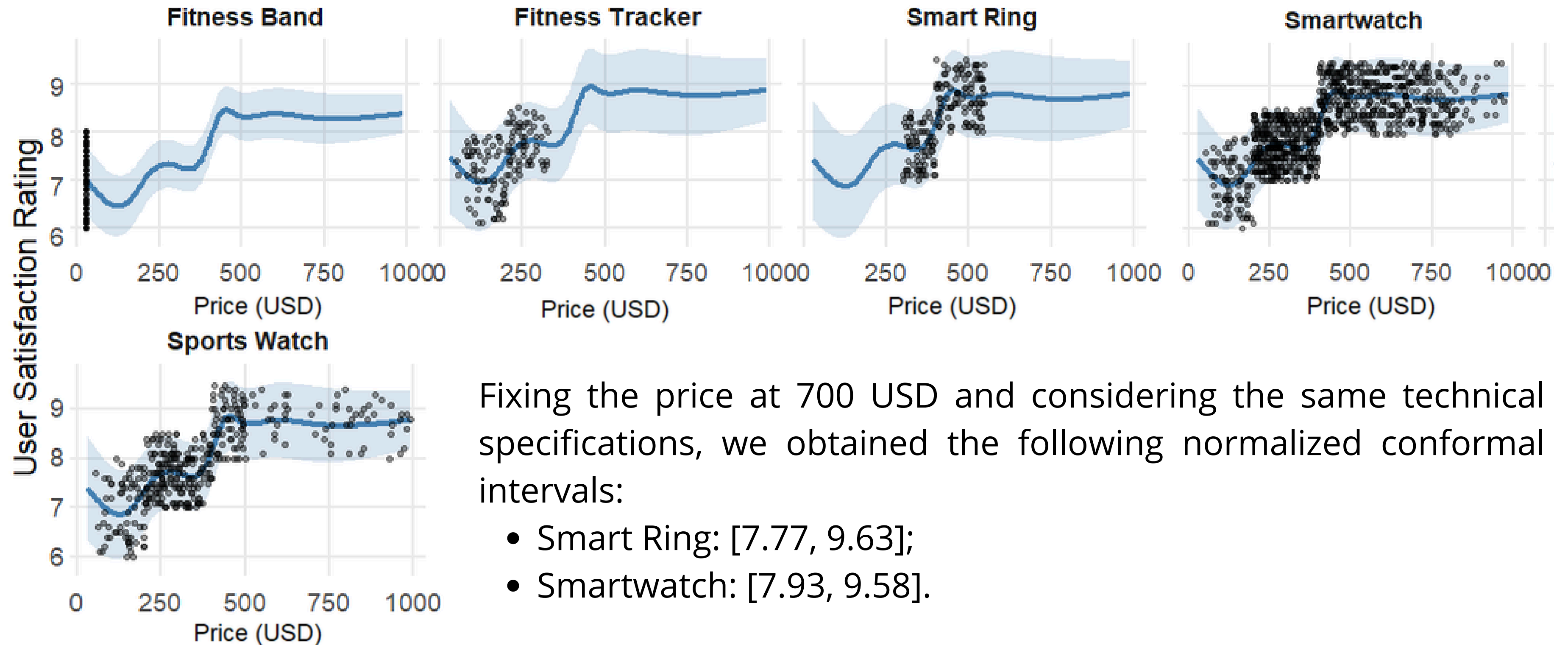
# RQ3: Predicting user satisfaction

After fitting the final GAM on the training set and computing the local scale parameter and the non-conformity scores we obtained the following **Normalized split conformal intervals**.



# RQ3: Predicting user satisfaction

## Normalized CP intervals by device category



# Conclusions and further developments

## Device Performance by Category

- Smartwatches: Best overall balance of accuracy, sensors, connectivity;
- Sports Watches: Prioritize battery life and robustness;
- Fitness Trackers/Bands: Simpler design and functionality.

## Price vs Performance

- Brand effects dominate pricing;
- Paying more  $\neq$  higher accuracy or functionality.

## Predicting User Satisfaction

- Satisfaction predicted using price and category;
- Conformal prediction provides distribution-free uncertainty intervals.

## Practical Implications

- Helps consumers select devices aligned with actual needs;
- Avoid overpaying for marginal technical gains.

## Future Directions

- Incorporate longitudinal usage and post-purchase behavior;
- Model brand perception over time for better satisfaction forecasts.

# Thank you for the attention!

## References

- **Kaggle**, *Wearable Health Device Performance Data 2025*. <https://www.kaggle.com/datasets/pratyushpuri/wearable-health-devices-performance-analysis>;
- **market.us**, Feb 2026. *Global Wearable Fitness Trackers Market*. <https://market.us/report/global-wearable-fitness-trackers-market/#overview>;
- **Rachael Schultz, Steven Cohen**, 2026. *The best fitness trackers and wearables of 2026*. <https://www.businessinsider.com/guides/tech/best-fitness-trackers#how-we-test>.